

SQL Study Notes

1. AND & OR Operator

- AND → requires all conditions true.
- OR → requires at least one condition true.
- Example: (Department='IT' AND Salary>80000) OR (Department='HR' AND Salary>60000).

2. LIKE Operator

- % matches multiple characters, _ matches one.
- Examples: LIKE 'A%', LIKE '%n', LIKE '%li%'.
- Industry: Retail(Laptops), Banking(Gmail), HR(Manager).

3. GROUP BY Operator

- Groups rows and applies aggregate functions.
- Example: SELECT Department, AVG(Salary) FROM Employees GROUP BY Department.

4. HAVING Clause

- Filters groups after aggregation.
- Example: HAVING AVG(Salary) > 80000.

5. CASE Clause

- Conditional logic (IF-THEN-ELSE).
- Example: CASE WHEN Salary>=80000 THEN 'High' ELSE 'Low' END.

6. Joins (INNER, LEFT, RIGHT)

- INNER JOIN → Only matches.
- LEFT JOIN → All left + matches.
- RIGHT JOIN → All right + matches.
- Example Scenario: Employees (Alice-IT, Bob-HR, Charlie-NULL); Departments (IT, HR, Finance).

7. Interview Questions (JOIN)

- Difference between INNER/OUTER JOIN.
- Self Join, Cross Join, USING vs ON.
- How to find unmatched records with LEFT JOIN.
- Performance & indexing considerations.

8. Quick Reference Cheat Sheet

- Basics: SHOW DATABASES; USE db; CREATE/DROP DATABASE;
- Tables: CREATE, DESCRIBE, DROP, ALTER;

- CRUD: INSERT, SELECT, UPDATE, DELETE;
- Filtering: WHERE, LIKE, IN, BETWEEN;
- Sorting: ORDER BY, LIMIT;
- Aggregates: COUNT, SUM, AVG, MIN, MAX;
- Grouping: GROUP BY, HAVING;
- Joins: INNER, LEFT, RIGHT, FULL;
- CASE, Subqueries, Constraints (PK, FK, UNIQUE, NOT NULL).